# Class 12: Genome Informatics

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### Section 1. Proportion of G/G in a Population

Imported a CSV file from Ensembl < https://www.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39894595-39895595;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel >

Here we will read in this CSV file:

```r
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")

head(mxl)
```

```
  Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
1                 NA19648 (F)                       A|A ALL, AMR, MXL      -
2                 NA19649 (M)                       G|G ALL, AMR, MXL      -
3                 NA19651 (F)                       A|A ALL, AMR, MXL      -
4                 NA19652 (M)                       G|G ALL, AMR, MXL      -
5                 NA19654 (F)                       G|G ALL, AMR, MXL      -
6                 NA19655 (M)                       A|G ALL, AMR, MXL      -
  Mother
1      -
2      -
3      -
4      -
5      -
6      -
```

```r
table(mxl$Genotype..forward.strand.)
```

```
A|A A|G G|A G|G 
 22  21  12   9 
```

> Question 5. What proportion of the Mexican Ancestry in Los Angeles sample population (MXL) are homozygous for the asthma associated SNP (G|G)?

```r
table(mxl$Genotype..forward.strand.) / nrow(mxl) * 100
```

```
    A|A     A|G     G|A     G|G 
34.3750 32.8125 18.7500 14.0625 
```

**14.0625%** of the Mexican Ancestry in Los Angeles sample population (MXL) are homozygous for the asthma associated SNP (G|G).

> Question 6. Now let's look at a different population, which is GBR (Great Britain), as stated in the lab sheet.

```r
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
gbr
```

```
   Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
1                  HG00096 (M)                       A|A ALL, EUR, GBR      -
2                  HG00097 (F)                       G|A ALL, EUR, GBR      -
3                  HG00099 (F)                       G|G ALL, EUR, GBR      -
4                  HG00100 (F)                       A|A ALL, EUR, GBR      -
5                  HG00101 (M)                       A|A ALL, EUR, GBR      -
6                  HG00102 (F)                       A|A ALL, EUR, GBR      -
7                  HG00103 (M)                       A|G ALL, EUR, GBR      -
8                  HG00105 (M)                       A|A ALL, EUR, GBR      -
9                  HG00106 (F)                       G|A ALL, EUR, GBR      -
10                 HG00107 (M)                       G|G ALL, EUR, GBR      -
11                 HG00108 (M)                       A|A ALL, EUR, GBR      -
12                 HG00109 (M)                       G|G ALL, EUR, GBR      -
13                 HG00110 (F)                       A|G ALL, EUR, GBR      -
14                 HG00111 (F)                       A|A ALL, EUR, GBR      -
15                 HG00112 (M)                       G|G ALL, EUR, GBR      -
```

| | | | | |
|---|---|---|---|---|
| 16 | HG00113 (M) | G\|G | ALL, EUR, GBR | - |
| 17 | HG00114 (M) | G\|A | ALL, EUR, GBR | - |
| 18 | HG00115 (M) | A\|G | ALL, EUR, GBR | - |
| 19 | HG00116 (M) | G\|G | ALL, EUR, GBR | - |
| 20 | HG00117 (M) | A\|A | ALL, EUR, GBR | - |
| 21 | HG00118 (F) | G\|G | ALL, EUR, GBR | - |
| 22 | HG00119 (M) | G\|A | ALL, EUR, GBR | - |
| 23 | HG00120 (F) | G\|G | ALL, EUR, GBR | - |
| 24 | HG00121 (F) | A\|G | ALL, EUR, GBR | - |
| 25 | HG00122 (F) | G\|G | ALL, EUR, GBR | - |
| 26 | HG00123 (F) | G\|A | ALL, EUR, GBR | - |
| 27 | HG00125 (F) | A\|G | ALL, EUR, GBR | - |
| 28 | HG00126 (M) | G\|G | ALL, EUR, GBR | - |
| 29 | HG00127 (F) | G\|A | ALL, EUR, GBR | - |
| 30 | HG00128 (F) | A\|G | ALL, EUR, GBR | - |
| 31 | HG00129 (M) | G\|G | ALL, EUR, GBR | - |
| 32 | HG00130 (F) | A\|G | ALL, EUR, GBR | - |
| 33 | HG00131 (M) | G\|G | ALL, EUR, GBR | - |
| 34 | HG00132 (F) | A\|A | ALL, EUR, GBR | - |
| 35 | HG00133 (F) | G\|A | ALL, EUR, GBR | - |
| 36 | HG00136 (M) | G\|G | ALL, EUR, GBR | - |
| 37 | HG00137 (F) | G\|A | ALL, EUR, GBR | - |
| 38 | HG00138 (M) | A\|A | ALL, EUR, GBR | - |
| 39 | HG00139 (M) | G\|G | ALL, EUR, GBR | - |
| 40 | HG00140 (M) | G\|A | ALL, EUR, GBR | - |
| 41 | HG00141 (M) | G\|G | ALL, EUR, GBR | - |
| 42 | HG00142 (M) | G\|G | ALL, EUR, GBR | - |
| 43 | HG00143 (M) | G\|A | ALL, EUR, GBR | - |
| 44 | HG00145 (M) | A\|A | ALL, EUR, GBR | - |
| 45 | HG00146 (F) | A\|A | ALL, EUR, GBR | - |
| 46 | HG00148 (M) | G\|A | ALL, EUR, GBR | - |
| 47 | HG00149 (M) | G\|A | ALL, EUR, GBR | - |
| 48 | HG00150 (F) | G\|A | ALL, EUR, GBR | - |
| 49 | HG00151 (M) | G\|A | ALL, EUR, GBR | - |
| 50 | HG00154 (F) | G\|G | ALL, EUR, GBR | - |
| 51 | HG00155 (M) | A\|G | ALL, EUR, GBR | - |
| 52 | HG00157 (M) | A\|A | ALL, EUR, GBR | - |
| 53 | HG00158 (F) | A\|A | ALL, EUR, GBR | - |
| 54 | HG00159 (M) | A\|A | ALL, EUR, GBR | - |
| 55 | HG00160 (M) | A\|A | ALL, EUR, GBR | - |
| 56 | HG00231 (F) | A\|G | ALL, EUR, GBR | - |
| 57 | HG00232 (F) | G\|G | ALL, EUR, GBR | - |
| 58 | HG00233 (F) | G\|G | ALL, EUR, GBR | - |

| | | | |
|---|---|---|---|
| 59 | HG00234 (M) | G\|G ALL, EUR, GBR | - |
| 60 | HG00235 (F) | A\|A ALL, EUR, GBR | - |
| 61 | HG00236 (F) | A\|A ALL, EUR, GBR | - |
| 62 | HG00237 (F) | A\|A ALL, EUR, GBR | - |
| 63 | HG00238 (F) | G\|G ALL, EUR, GBR | - |
| 64 | HG00239 (F) | G\|A ALL, EUR, GBR | - |
| 65 | HG00240 (F) | G\|A ALL, EUR, GBR | - |
| 66 | HG00242 (M) | G\|A ALL, EUR, GBR | - |
| 67 | HG00243 (M) | A\|G ALL, EUR, GBR | - |
| 68 | HG00244 (M) | G\|A ALL, EUR, GBR | - |
| 69 | HG00245 (F) | A\|G ALL, EUR, GBR | - |
| 70 | HG00246 (M) | A\|G ALL, EUR, GBR | - |
| 71 | HG00250 (F) | G\|G ALL, EUR, GBR | - |
| 72 | HG00251 (M) | G\|A ALL, EUR, GBR | - |
| 73 | HG00252 (M) | G\|A ALL, EUR, GBR | - |
| 74 | HG00253 (F) | A\|A ALL, EUR, GBR | - |
| 75 | HG00254 (F) | A\|G ALL, EUR, GBR | - |
| 76 | HG00255 (F) | A\|G ALL, EUR, GBR | - |
| 77 | HG00256 (M) | A\|G ALL, EUR, GBR | - |
| 78 | HG00257 (F) | G\|G ALL, EUR, GBR | - |
| 79 | HG00258 (F) | A\|A ALL, EUR, GBR | - |
| 80 | HG00259 (F) | G\|A ALL, EUR, GBR | - |
| 81 | HG00260 (M) | G\|G ALL, EUR, GBR | - |
| 82 | HG00261 (F) | G\|G ALL, EUR, GBR | - |
| 83 | HG00262 (F) | A\|A ALL, EUR, GBR | - |
| 84 | HG00263 (F) | G\|A ALL, EUR, GBR | - |
| 85 | HG00264 (M) | A\|G ALL, EUR, GBR | - |
| 86 | HG00265 (M) | G\|G ALL, EUR, GBR | - |
| 87 | HG01334 (M) | A\|G ALL, EUR, GBR | - |
| 88 | HG01789 (M) | G\|A ALL, EUR, GBR | - |
| 89 | HG01790 (F) | G\|A ALL, EUR, GBR | - |
| 90 | HG01791 (M) | A\|A ALL, EUR, GBR | - |
| 91 | HG02215 (F) | G\|G ALL, EUR, GBR | - |

| | Mother |
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| 89 | – |
| 90 | – |
| 91 | – |

Find proportion of G|G

```r
round(table(gbr$Genotype..forward.strand.) / nrow(gbr) * 100, 2)
```

```

  A|A   A|G   G|A   G|G 
25.27 18.68 26.37 29.67 
```

Based on these results, it can be indicated that this variant, which is associated with childhood asthma, is more frequently found in the GBR population than in the MXL population.

## Section 4: Population Scale Analysis [HOMEWORK]

> Question 13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

*Hint: The read.table(), summary() and boxplot() functions will likely be useful here. There is an example R script online to be used ONLY if you are struggling in vein. Note that you can find the medium value from saving the output of the boxplot() function to an R object and examining this object. There is also the medium() and summary() function that you can use to check your understanding.*

```r
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

```
   sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

**Total**

```r
nrow(expr)
```

```
[1] 462
```

**Sample Sizes for Each Genotype**

```r
table(expr$geno)
```

```

A/A A/G G/G 
108 233 121 
```

### Median Expression Levels

```r
colnames(expr)
```

```
[1] "sample" "geno"   "exp"   
```

```r
tapply(expr$exp, expr$geno, median)
```

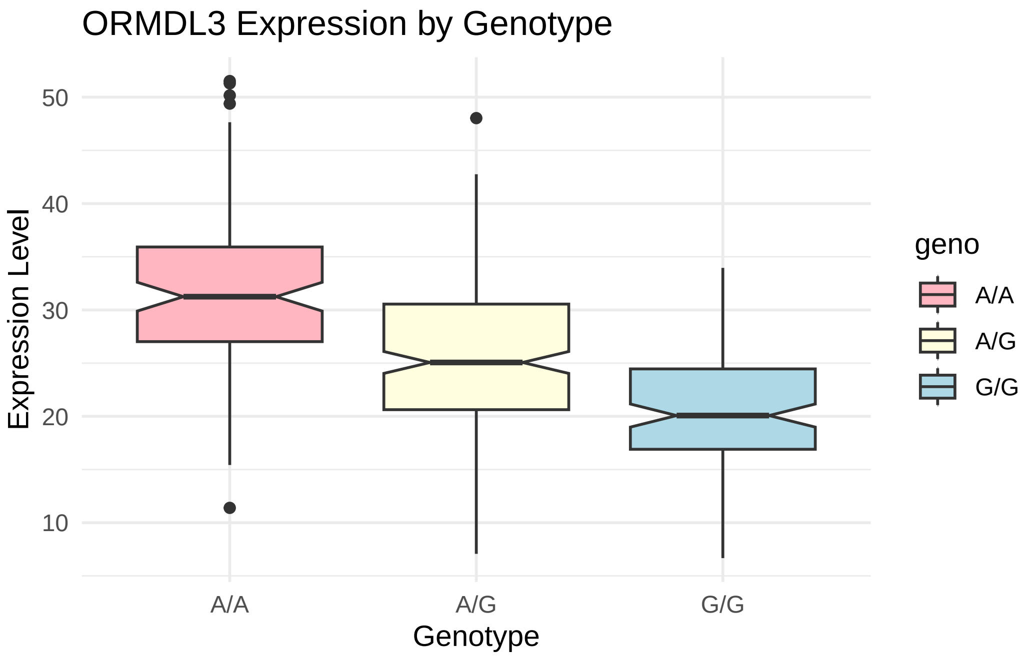
```
     A/A      A/G      G/G 
31.24847 25.06486 20.07363 
```

These results show that there are 233 A|G genotypes, 108 A|A genotypes, and 121 G|G genotypes, which makes for a total of 462 genotypes. The median expression level is 31.248 for A|A, 25.064 for A|G, and 20.074 for G|G.

> Question 14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```r
library(ggplot2)
```

```r
ggplot(expr) +
  aes(x = geno, y = exp, fill = geno) +
  geom_boxplot(notch = TRUE) +
  scale_fill_manual(values = c(
    "A/A" = "lightpink",
    "A/G" = "lightyellow",
    "G/G" = "lightblue"
  )) +
  labs(
    title = "ORMDL3 Expression by Genotype",
    x = "Genotype",
    y = "Expression Level"
  ) +
  theme_minimal()
```




Analysis: The boxplot illustrates the distribution of ORMDL3 gene expression levels across the three genotypes: A/A, A/G, and G/G. The A/A genotype shows the highest median expression, A/G displays intermediate expression, and G/G has the lowest median expression, suggesting that the presence of the G allele is associated with progressively reduced ORMDL3 expression. Differences in spread and outliers among the groups indicate variation within each genotype, but the overall downward trend supports a potential functional effect of this SNP on gene regulation and expression levels.